Applications

1 An investment scheme advertises the following percentage returns after 2 years based on historical probabilities:

Percentage return (x)	10%	15%	25%
$P(X = x)$	0.7	0.15	0.15

a Find the expected percentage return on an investment.

b Find how much an investment of $\$50\,000$ is expected to be worth after 2 years.

2 A salesperson is starting work in a new region and analyses the probability of how many sales he is likely to make in the next month:

Number of sales (x)	0	1	2	3	4
$P(X = x)$	0.35	0.25	0.2	0.15	0.05

a Given that he makes at least one sale, state the probability that he will make 2 sales.

b The salesperson is offered 2 payment schemes:

- Option A: Flat monthly income of $\$1800$
- Option B: $\$1000$ flat fee per month plus $\$500$ per sale

If he chooses option B, what is his expected monthly income?

c Which option should he choose to maximise his income?

3 At a car park in the city, all day parking is charged on the following basis below:

- Cars with just a driver pay $\$25$
- Cars with a driver and one passenger pay $\$15$
- Cars with a driver and at least two passengers pay $\$12$

The number of people in one of these cars on a given day is summarised in the table:

Number of people	1	2	3	4	5
Number of cars	4500	3500	1100	600	300

a Find the probability a randomly selected car is carrying 3 people.

b Given that a car was carrying at least 2 people, find the probability it was carrying 4.

c Let X represent the parking fee paid by a randomly selected car. Construct the probability distribution table for X.

d Find the expected revenue per car in this car park.

e Find the standard deviation of the reve[illegible]er car in this car park.

4 To be allowed to leave class and go to lunch, Mrs Ammar gives her class a 3-question multiple choice quiz, with each question consisting of 4 possible answers. David hasn't listened to a word Mrs Ammar has said all lesson and will have to guess each of the three questions.

a Find the probability David will guess none of the questions correct.

b Find the probability David will guess all of the questions correctly.

c Find the probability David will guess just one of the questions correctly.

d Let X represent the number of questions David guesses correctly. Construct a probability distribution table for X.

e State whether the table represents a discrete probability distribution. Explain your answer.

5 When a student completes a task set by their teacher on Spacemaths, the number of hints used is monitored by the system:

- The probability of using at least 1 hint is 0.6.
- The probability of using 2 hints is the same as using 3 hints.
- The probability of using 1 hint is the same as using 4 hints.
- At most students can use 4 hints.
- The probability they use 2 hints is half the probability that they use 0 hints.

a Let X represent the number of hints they used. Construct a probability distribution table for X.

b Find the expected number of hints a student will use.

c Given that a student used at least 2 hints, find the probability they used 4 hints.

6 Jo and Ky are playing a game of cards that is either won or lost, there is no draw. The probability that Jo wins the first game is 0.6.

- If Jo wins a game, the probability he wins the next game is 0.7.
- If Jo loses a game, the probability that Ky wins the next game is 0.8.
- They keep playing until either Jo or Ky wins two games.

a Construct a probability tree diagram of this situation.

b Let X represent the number of games of cards played before someone wins two games. Construct a probability distribution table for X.

c Find $E(X)$.

d Given that Jo won, calculate the probability that 3 games were played.

7 A fair standard die is thrown. Let X be the [redacted]ber of dots on the uppermost face.

- **a** Construct the probability distribution table for X.
- **b** Is the discrete probability distribution uniform or non-uniform?
- **c** Find $E\left(X\right)$.
- **d** Find $\text{Var}\left(X\right)$.

8 Two normal dice are rolled and the sum of the numbers on the uppermost face recorded. Let Y represent the value of the sum of the two dice.

- **a** Complete the table for this discrete probability distribution:

y	2	3	4	5	6	7	8	9	10	11	12
$P(Y=y)$	$\dfrac{1}{36}$		$\dfrac{3}{36}$			$\dfrac{6}{36}$					$\dfrac{1}{36}$

- **b** State the most likely sum to occur.
- **c** Describe the shape of the distribution.
- **d** Find the expected value.
- **e** Find the variance.

9 A die was manufactured such that an odd number is twice as likely to be rolled as an even number. Let X be the number the die lands on.

- **a** Construct a probability distribution table for X.
- **b** Find $E\left(X\right)$.
- **c** Find $\text{Var}\left(X\right)$.

10 A game is played in which a standard six-sided die is rolled. If it lands on a number other than 1, then the score is that number. If it lands on 1, then a second four-sided dice with numbers 3 to 6 is rolled and the number that die lands on is the score. Let X be the score of a player in this game.

- **a** Construct a probability distribution table for X.
- **b** Find $E\left(X\right)$.
- **c** Find $\text{Var}\left(X\right)$.

11 A six-sided die with numbers from 1 to 6 is weighted such that $P(\text{ prime number }) = 0.1$ and $P(4) = P(6) = 0.3$. Let X represent the possible outcomes from one roll of the dice.

a Construct the probability distribution table for X.

b Find the following:

i $P(X < 3)$

ii $P(X = 3 | X \leq 5)$

iii $P(X < 3 | X < 5)$

iv $P(X < 4 | X \geq 2)$

12 A fair standard die is thrown onto the ground and the number of visible odd-numbered faces (the faces which are not on the ground) is noted. Let Y be the number of visible odd-numbered faces.

a Construct the probability distribution for Y.

b Is this discrete probability distribution uniform or non-uniform?

13 A fair standard die is rolled and the number of dots on the visible faces (the faces which are not on the ground) is noted. Let W be the number of dots that can be seen on the visible faces.

a Construct the probability distribution table for W.

b Is the discrete probability distribution uniform or non-uniform?

14 A regular six-sided dice has a side length of 8 cm. The dice is rolled on the ground and the height above ground of the dot on the face with only a single dot is noted. Let H be the number of centimetres this single dot is above the ground.

a List the possible outcomes for H.

b Hence, construct the probability distribution table for X.

15 Two dice are rolled and the absolute value of the differences between the numbers appearing uppermost are recorded.

a Complete the sample space in the given table.

b Let X be defined as the absolute value of the difference between the two dice. Construct the probability distribution table for X.

c Find $P(X < 3)$

d Find $P(X \leq 4 | X \geq 2)$

	1	2	3	4	5	6
1	0			3		
2	1					
3					2	
4						
5	4		2			
6						

16 Two dice are rolled and the difference between the largest number and smallest number is calculated. A player wins $\$1$ if the difference is 3, $\$2$ if the difference is 4 , $\$3$ if the difference is 5 and $\$0$ otherwise.

a Complete the sample space in the given table.

b Let X be the winnings from one game. Construct a probability distribution table for X.

c Find the expected winnings.

d If it costs $\$2$ to play each game, find the player's expected return.

	1	2	3	4	5	6
1	0	1	2			
2						
3						
4						
5						
6						

17 At a local fair, in a game that involves rolling a standard six-sided die, players can win a prize depending on what they roll. Each player must pay $\$3$ to play. The prizes are awarded as follows:

- The player wins $\$3$ if a 1, 3 or 5 is rolled.
- The player wins $\$6$ if a 4 or 6 is rolled.
- The player wins $\$9$ if a 2 is rolled.

a Let X be the prize received by the player. Construct a probability distribution table for X .

b Find the expected prize value.

c Find the standard deviation of the distribution.

18 At a fair, a games stall operator offers prizes worth $\$1.50$, $\$2$, $\$1$ and $\$0.50$ for one attempt at a particular game. The probabilities of winning these prizes are respectively 0.15, 0.01, 0.05 and 0.03.

a Find the probability of not winning a prize.

b If each game costs $\$2$, find the expected profit per game for the operator in dollars.

c If each game costs C and the games stall operator made a profit of $\$595$ from 500 games. Find C, the amount he likely charged per game in dollars.

19 The probability that a particular biased coin lands on tails is 0.7. Let X be the number of tails when the coin is tossed twice. Complete the given probability distribution table for X.

x	0	1	2
$P(X=x)$			

20 In a game of two-up, a person called the “[redacted]er” tosses two coins:

- If the coins land with two heads up, then the Spinner wins and the gamblers lose.
- If the coins land with two tails up, the Spinner loses and the gamblers win.
- If the coins land one head up and one tail up, the Spinner tosses the coins again and the gamblers break even.

a Construct a tree diagram to represent all possible outcomes of tossing two coins.

b If each gambler bets $\$3$, and can win $\$3$ per toss, construct a probability distribution table for the profit of the gambler for one game of two-up.

21 An unfair coin is tossed. The chance of tails facing upwards after the toss is 30%.

a Find the probability of the coin landing tails up for the first time on the third toss.

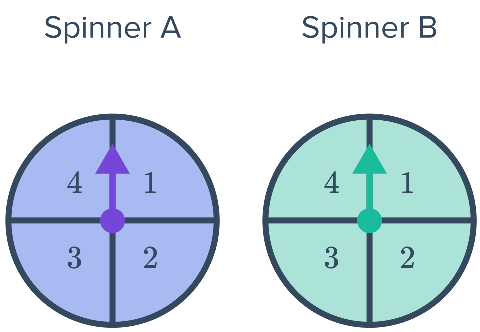
b Find the probability of the coin landing tails up for the first time on the fourth toss.

c Find the probability that it takes four tosses of the coin before you see a tail on the fifth toss.

d Let N be the number of tosses of the coin it takes before you see a tail on the next toss. Define the probability density function for N.

22 Two fair spinners, A and B, are spun. The number from each spinner is noted and the total score is defined below:

$$X = \begin{cases} A+B; \text{ if } A=B \\ A+B; \text{ if } A>B \\ B-A; \text{ if } A<B \end{cases}$$



a List all the possible outcomes of X.

b Construct a probability distribution for X.

23 A spinner has four sections each numbere  1 to 4. The spinner is divided according to the given equations. Let X represent the number spun on the spinner.

- $P(1) = P(2) + P(3) + P(4)$
- $P(2) = 2P(3)$
- $P(3) = P(4)$

a Construct the probability distribution table for X.

b Hence, construct the cumulative probability distribution table for X.

c Find the following:

i $P(X < 3)$.

ii $P(X \geq 3)$.

iii $P(X = 1 \cup X = 3)$.

iv $P(X \leq 3 | X > 1)$.

24 Two spinners numbered from 0 to 4 are spun. Let X be the product of the two numbers that come up.

a List all the possible outcomes of X.

b Construct a probability distribution table for X.

c Find $E(X)$.

d Find $\mathrm{Var}(X)$.

25 Three marbles are randomly drawn from a bag containing seven black and three green marbles. Let X be the number of black marbles drawn.

a Construct the probability distribution table for X if the marbles are drawn with replacement.

b Construct the probability distribution table for X if the marbles are drawn without replacement.

26 In Brad's toy box, there are 3 toy cars and 4 toy dinosaurs. Each day, for three days, he takes a toy at random and plays with it, and then puts it back.

a Construct a probability tree diagram of all the possible combination of toys he could have played with over these three days.

b Let X be the number of days he played with a toy car. Construct the probability distribution table for the discrete random variable X.

c Find the expected number of days he will play with the toy car.

d Find the standard deviation for the distribution of X.

27 A pencil case contains 6 blue pens and 5 [redacted] pens. 4 pens are drawn randomly from the pencil case without replacement.

- **a** Find the probability of drawing one blue pen from the pencil case.
- **b** Find the probability of drawing three blue pens from the pencil case.
- **c** Let X be the number of blue pens drawn. Complete the probability distribution table:

x	0	1	2	3	4
$P(X = x)$	$\dfrac{1}{66}$		$\dfrac{5}{11}$		

28 A pencil case contains 9 blue pens and 5 green pens. 4 pens are drawn randomly from the pencil case, one at a time, each being replaced before the next one is drawn.

- **a** Find the probability of drawing one blue pen from the pencil case.
- **b** Find the probability of drawing three blue pens from the pencil case.
- **c** Let X be the number of blue pens drawn. Complete the probability distribution table:

x	0	1	2	3	4
$P(X = x)$	$\dfrac{625}{38\,416}$		$\dfrac{6075}{19\,208}$		

29 Two earrings are taken without replacement from a draw containing 3 black earrings and 5 brown earrings. Let X be the number of black earrings drawn.

- **a** Construct a probability distribution for X.
- **b** Given that at least one black earring was selected, find the probability that two were selected.

30 A child randomly selects 3 balls without replacement from a box containing 8 red balls and 2 green balls. For every red ball chosen, the child receives 1 chocolate. For every green ball chosen, the child receives 5 chocolates.

- **a** Let X be the number of green balls chosen. Construct the probability distribution table for X.
- **b** Let T be the number of chocolates the child receives. Construct the probability distribution table for T.
- **c** State the most likely number of chocolates that the child receives.
- **d** State the expected number of chocolates that the child receives.